ELORALINTIDE

JOYOUS RESEARCH GUIDE

LY3841136 | Selective Amylin Receptor Agonist

Research Category: Metabolic Biology
Primary Research Axis: Amylin Signaling • Satiety • Energy Balance • Weight Regulation
Development Status: Phase 3 Clinical Development
Developer: Eli Lilly and Company
Administration Studied: Once-weekly subcutaneous administration
Regulatory Status: Investigational / Not FDA Approved

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01 | COMPOUND IDENTITY

Compound

Eloralintide

Development Code

LY3841136

Biological Classification

Long-acting selective amylin receptor agonist

Primary Molecular Target

Amylin receptor system

Primary Research Areas

- Satiety signaling
- Food intake regulation
- Energy balance
- Body-weight regulation
- Obesity biology
- Amylin and incretin interaction

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02 | QUICK SUMMARY

Eloralintide is an investigational once-weekly compound being developed by Eli Lilly for obesity and related metabolic conditions.

What makes it particularly interesting is **how it works**.

Most of the metabolic medications familiar to the public today center on incretin biology, especially GLP-1 and GIP.

Eloralintide instead targets the **amylin receptor system**.

Amylin is a naturally occurring hormone released from pancreatic beta cells alongside insulin after eating. Among its physiological roles, amylin participates in signaling related to satiation and food intake.

In everyday language:

GLP-1 is not the only biological system capable of telling the brain that enough food has been consumed. Amylin is another important part of that conversation.

Eloralintide is designed to selectively activate that pathway.

That makes it important both as a potential stand-alone metabolic therapy and as a possible future partner to incretin-based therapy.

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03 | WHY RESEARCHERS ARE INTERESTED

The first generation of modern obesity medications demonstrated that manipulating appetite and metabolic signaling can produce substantial changes in body weight.

The next question is bigger:

Can different biological pathways be targeted independently or together to improve efficacy, tolerability, or both?

Eloralintide gives researchers another major pathway to investigate.

Instead of simply creating another GLP-1 receptor agonist, Lilly is exploring selective amylin receptor activation.

That opens several important research questions:

Can amylin signaling independently produce substantial weight reduction?

Phase 2 suggests that it can.

Is there a meaningful dose-response relationship?

Again, Phase 2 suggests that there is.

Does the way someone reaches a maintenance dose influence tolerability?

The Phase 2 trial specifically investigated this question using different escalation strategies.

Could amylin complement incretin therapy?

Lilly is now directly investigating this concept in Phase 3.

This may ultimately become one of the most important parts of the eloralintide story.

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04 | MECHANISM OVERVIEW

The Amylin Pathway

Amylin is a peptide hormone normally co-secreted with insulin from pancreatic beta cells following food intake.

Its physiological signaling contributes to several processes associated with eating and energy regulation.

These include:

SATIATION SIGNALING

Helping communicate that sufficient food has been consumed.

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MEAL TERMINATION

Contributing to the biological processes that influence when eating stops.

↓

FOOD INTAKE

Reduced caloric intake can follow stronger satiation signaling.

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ENERGY BALANCE

Repeated changes in food intake can influence long-term energy balance.

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BODY WEIGHT

Over time, sustained changes in energy intake can contribute to changes in body weight.

Simplified Biological Systems Map

Food Intake

↓

Pancreatic Amylin Signaling

↓

Amylin Receptors

↓

Central Satiety Networks

↓

Earlier / Stronger Satiation

↓

Reduced Energy Intake

↓

Changes in Body-Weight Regulation

This is intentionally simplified. Amylin biology involves interconnected peripheral and central signaling systems rather than one simple linear pathway.

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05 | ELORALINTIDE VS. GLP-1

Eloralintide and GLP-1 receptor agonists can both influence eating behavior and body weight, but they do not primarily act through the same receptor.

	Eloralintide	GLP-1 Therapy
Primary pathway	Amylin	GLP-1
Primary receptor target	Amylin receptors	GLP-1 receptor
Satiety effects	Yes	Yes
Food-intake effects	Yes	Yes
Glucose-related effects	Indirect / pathway dependent	Major component
Once-weekly development	Yes	Common
Potential combination research	Yes	Yes

The important lesson is:

Similar outcomes do not necessarily mean identical biology.

Two therapies may both reduce appetite or body weight while reaching that result through different signaling pathways.

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06 | CLINICAL DEVELOPMENT SNAPSHOT

Phase 1

Early human research established initial pharmacology, exposure, safety, tolerability, and biological activity across a range of doses.

Phase 2

Researchers tested multiple once-weekly doses and escalation strategies for 48 weeks in adults with obesity or overweight without type 2 diabetes.

Phase 3

The ENLIGHTEN clinical program is now evaluating eloralintide in substantially larger and more diverse obesity populations.

Current Evidence Position

Promising human Phase 2 evidence

with

Phase 3 confirmation pending.

That distinction is important.

Eloralintide has progressed considerably beyond preclinical or proof-of-concept research, but it does not yet have an established FDA-approved treatment protocol.

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07 | THE PHASE 2 STUDY

The major Phase 2 obesity study was a:

48-week

randomized

double-blind

placebo-controlled

multicenter clinical trial.

A total of **263 adults** were randomized across 46 research centers in the United States.

Participants were 18 to 75 years old and had either:

- BMI $\geq$30 kg/m^2

or

- BMI $\geq$27 kg/m^2 with at least one weight-related comorbidity.

Importantly:

Participants with type 2 diabetes were excluded.

This matters when interpreting the results because Phase 2 does not tell us whether exactly the same outcomes should be expected in people with type 2 diabetes.

Phase 3 is addressing that question separately.

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08 | PHASE 2 DOSE MAP

Researchers did not test just one eloralintide dose.

They deliberately tested several different exposures.

Fixed-Dose Arms

1 mg

1 mg once weekly for 48 weeks

Randomized participants: 28

Mean body-weight change at Week 48:

approximately −9%

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3 mg

3 mg once weekly for 48 weeks

Randomized participants: 24

Mean body-weight change:

approximately −12%

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6 mg

6 mg once weekly for 48 weeks

Randomized participants: 28

Mean body-weight change:

approximately −18%

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9 mg

9 mg once weekly for 48 weeks

Randomized participants: 54

Mean body-weight change:

approximately −20%

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09 | PHASE 2 TITRATION ARMS

This is where the study becomes especially interesting.

Researchers also tested two different paths toward 9 mg.

ROAD 1

6 mg → 9 mg

Weeks 1–20

6 mg once weekly

↓

Weeks 21–48

9 mg once weekly

Randomized participants: 24

Mean body-weight change at Week 48:

approximately −20%

ROAD 2

3 mg → 6 mg → 9 mg

Weeks 1–4

3 mg once weekly

↓

Weeks 5–8

6 mg once weekly

↓

Weeks 9–48

9 mg once weekly

Randomized participants: 52

Mean body-weight change at Week 48:

approximately −16%

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10 | PHASE 2 WEEKLY PROTOCOL MAP

Weeks 1–4

1 mg Fixed → **1 mg**

3 mg Fixed → **3 mg**

6 mg Fixed → **6 mg**

9 mg Fixed → **9 mg**

6→9 Arm → **6 mg**

3→6→9 Arm → **3 mg**

Weeks 5–8

1 mg Fixed → **1 mg**

3 mg Fixed → **3 mg**

6 mg Fixed → **6 mg**

9 mg Fixed → **9 mg**

6→9 Arm → **6 mg**

3→6→9 Arm → **6 mg**

Weeks 9–20

1 mg Fixed → **1 mg**

3 mg Fixed → **3 mg**

6 mg Fixed → **6 mg**

9 mg Fixed → **9 mg**

6→9 Arm → **6 mg**

3→6→9 Arm → **9 mg**

Weeks 21–48

1 mg Fixed → **1 mg**

3 mg Fixed → **3 mg**

6 mg Fixed → **6 mg**

9 mg Fixed → **9 mg**

6→9 Arm → **9 mg**

3→6→9 Arm → **9 mg**

RESEARCH CONTEXT

These are clinical-trial arms.

They document **what researchers studied**, not how an individual should use eloralintide.

There is currently no FDA-approved eloralintide dose or titration schedule.

11 | WHY WERE THERE THREE ROADS TO 9 MG?

This is one of the most useful lessons in the entire eloralintide study.

Researchers effectively investigated three different experiences involving 9 mg.

DIRECT

9 mg

Start at 9 mg and remain there.

LONG LEAD-IN

6 mg → 9 mg

Remain at 6 mg for 20 weeks before advancing to 9 mg.

GRADUAL ESCALATION

3 mg → 6 mg → 9 mg

Increase through two lower exposure levels before reaching 9 mg.

Why do this?

Because researchers were not simply asking:

"Does 9 mg work?"

They were also asking:

"Does how you get to 9 mg matter?"

12 | WHY TITRATION ARMS EXIST

Titration is frequently misunderstood.

Seeing an escalation arm in a clinical trial does not automatically mean:

"This is the recommended dosing schedule."

Dose-ranging studies use escalation strategies to answer research questions.

QUESTION 1

Does gradual exposure improve tolerability?

A biologically effective dose may produce more adverse effects when introduced rapidly.

Researchers can test whether lower initial exposure changes that experience.

QUESTION 2

How much activity occurs at lower doses?

The 1 mg and 3 mg groups were not simply stepping stones.

They were independent research arms.

Both produced measurable average weight reduction.

This helps establish the drug's **dose-response curve**.

QUESTION 3

Does escalation improve treatment persistence?

Maximum biological activity is not the only goal.

A therapy also has to be tolerable enough for participants to remain on treatment.

Researchers therefore examine:

- adverse events
- dose interruptions
- discontinuations
- tolerability
- treatment persistence

QUESTION 4

Does the route to maintenance dose change the outcome?

Direct 9 mg and 6→9 mg produced similar mean Week 48 weight reductions.

The 3→6→9 mg group produced a smaller mean reduction.

But this does **not** prove that slower titration causes less weight loss.

These were relatively small Phase 2 groups, and multiple factors can influence group-level outcomes.

That is exactly why Phase 3 exists.

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13 | WHAT PHASE 2 ACTUALLY SHOWED

Mean Body-Weight Change at Week 48

Research Arm	Mean Change
Placebo	−0.4%
1 mg	≈ −9%
3 mg	≈ −12%
6 mg	≈ −18%
9 mg	≈ −20%
6→9 mg	≈ −20%
3→6→9 mg	≈ −16%

The overall pattern suggests a meaningful **dose-response relationship**.

Higher-dose strategies generally produced greater average weight reduction than lower doses.

But another piece of the story is equally important:

Efficacy and tolerability have to be considered together.

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14 | TOLERABILITY

The two most frequently reported adverse events in the Phase 2 publication were **nausea and fatigue**.

Reported Nausea

1 mg → **11%**

3 mg → **13%**

6 mg → **64%**

9 mg → **33%**

6→9 mg → **54%**

3→6→9 mg → **25%**

Placebo → **14%**

Reported Fatigue

1 mg → **0%**

3 mg → **13%**

6 mg → **29%**

9 mg → **43%**

6→9 mg → **46%**

3→6→9 mg → **21%**

Placebo → **12%**

Notice something important.

The percentages do not increase in a perfectly straight line as the dose rises.

For example:

6 mg produced more reported nausea than 9 mg.

That does not mean 9 mg is inherently easier to tolerate than 6 mg.

Phase 2 groups were relatively small, and percentages can vary considerably between groups.

JOYOUS RESEARCH NOTE

Never read one adverse-event percentage in isolation.

Look at:

dose + group size + escalation strategy + duration + discontinuations + overall safety pattern.

That is how clinical-trial data should be interpreted.

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15 | THE MOST IMPORTANT PHASE 2 TAKEAWAY

It would be easy to summarize this study as:

"9 mg produced about 20% weight loss."

But that misses most of what makes the trial valuable.

The better takeaway is:

Eloralintide demonstrated substantial biological activity across a range of doses, while different dose and escalation strategies produced different efficacy and tolerability experiences.

That is precisely the information researchers need before designing Phase 3.

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16 | WHAT PHASE 3 CHANGES ABOUT DOSING

Phase 2 builds the map.

Phase 3 determines which parts of that map survive much larger testing.

The Phase 2 study gives us the clearest publicly established milligram dose framework:

1 mg | 3 mg | 6 mg | 9 mg

But those doses should not automatically be interpreted as future commercial doses.

Phase 3 can refine:

- starting dose
- escalation sequence
- escalation interval
- maintenance dose
- maximum dose
- dose-hold criteria
- tolerability adjustments
- population-specific strategies

Public Phase 3 registry information does not currently provide a final FDA-approved dosing schedule.

Until detailed Phase 3 results and regulatory labeling become available:

The 1–9 mg framework belongs to the Phase 2 research story.

It is not an established clinical dosing range.

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17 | THE ENLIGHTEN PHASE 3 PROGRAM

Lilly has moved eloralintide into a broad Phase 3 development program called **ENLIGHTEN**.

This is important because researchers are no longer asking only whether eloralintide can reduce weight in one population.

They are testing where amylin therapy may fit across obesity medicine.

ENLIGHTEN-1

Obesity / Overweight Without Type 2 Diabetes

This is the large Phase 3 population most closely related to the original Phase 2 study.

Planned enrollment:

approximately 1,980 participants

The study includes an extended follow-up period for participants with prediabetes.

Research Question

Can the efficacy and safety observed in Phase 2 be reproduced in a much larger population?

ENLIGHTEN-2

Obesity / Overweight With Type 2 Diabetes

Planned enrollment:

approximately 1,035 participants

This fills an important evidence gap because participants with type 2 diabetes were excluded from the major Phase 2 obesity trial.

Research Question

How does eloralintide perform when obesity and type 2 diabetes occur together?

ENLIGHTEN-3

Obesity / Overweight With Obstructive Sleep Apnea

This expands the research beyond body weight alone.

Research Question

Can metabolic improvement and weight reduction meaningfully affect an obesity-associated condition such as obstructive sleep apnea?

ENLIGHTEN-4

Obesity / Overweight With Knee Osteoarthritis Pain

This study investigates another important consequence of obesity.

Research Question

Can substantial weight reduction translate into meaningful improvement in obesity-associated knee osteoarthritis outcomes?

ENLIGHTEN-6

Persistent Obesity on Background Weekly Incretin Therapy

This may become one of the most interesting trials in the entire program.

Participants enter the study while being treated with a weekly incretin.

Researchers can therefore investigate whether adding selective amylin signaling provides additional benefit.

Research Question

What happens when incretin biology alone is not enough?

This begins moving the conversation from:

AMYLIN vs. INCRETIN

toward:

AMYLIN + INCRETIN

That distinction could become extremely important in the next generation of metabolic therapy.

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18 | WHAT PHASE 3 CAN TELL US THAT PHASE 2 CANNOT

Phase 3 gives researchers substantially more statistical power and broader clinical exposure.

Important questions include:

EFFICACY

Does approximately 18–20% mean weight reduction reproduce in larger populations?

DOSE OPTIMIZATION

Which exposure strategy gives the strongest balance between efficacy and tolerability?

DIABETES

Does type 2 diabetes alter the response?

COMBINATION BIOLOGY

Can amylin provide additional benefit when used alongside established incretin therapy?

OBESITY COMPLICATIONS

Does weight reduction translate into improvements beyond the scale?

SAFETY

What does the adverse-event profile look like when thousands rather than hundreds of participants are exposed?

DURABILITY

Does weight reduction continue, plateau, or change with longer treatment?

These are the questions that begin transforming an interesting investigational molecule into a potential clinical therapy.

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19 | HOW TO READ FUTURE ELORALINTIDE HEADLINES

As Phase 3 data begin appearing, avoid looking only at the largest weight-loss percentage.

Ask:

What population was studied?

With diabetes?

Without diabetes?

Already using an incretin?

What dose strategy was used?

Starting dose and maintenance dose both matter.

What was the duration?

Twenty-four weeks and seventy-two weeks are not interchangeable.

What estimand is being reported?

Clinical trials may report results using different assumptions about discontinuation and adherence.

How many participants discontinued?

Efficacy matters.

So does whether people can remain on therapy.

What happened to lean mass?

Weight loss and fat loss are not identical measurements.

What happened to cardiometabolic markers?

Look beyond the scale.

What were the adverse events?

Especially gastrointestinal effects and fatigue.

This is how a headline becomes useful clinical-trial information.

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20 | WEEKLY METABOLIC RESEARCH LOG

When following an approved clinician-guided metabolic therapy, structured tracking can make trends easier to interpret.

DOSE

Record:

- exact weekly dose
- administration date
- escalation date
- held or delayed doses

NAUSEA

Track symptoms for approximately 48–72 hours after administration.

A simple 0–10 scale makes week-to-week comparisons easier.

NUTRITION

Record:

- approximate meal size
- protein intake
- significant appetite changes

Reduced appetite does not automatically mean adequate nutrition.

GI + HYDRATION

Track:

- bowel pattern
- constipation
- diarrhea
- vomiting
- hydration

BODY COMPOSITION SIGNALS

Track:

- weekly average weight
- waist measurement

Weekly averages are generally more useful for identifying trends than individual daily scale readings.

DOSE CHANGES

When a dose is held or reduced, record why.

Examples include:

- adverse effects
- illness
- poor oral intake
- dehydration
- procedure
- clinician instruction

- medication change

Context makes the data useful.

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21 | THE THREE-QUESTION TROUBLESHOOTING FRAMEWORK

When someone using an approved clinician-guided metabolic therapy feels that the expected response is not occurring, three very different problems can look similar.

QUESTION 1

Is exposure insufficient?

The dose or duration may not yet be producing the expected biological response.

QUESTION 2

Was exposure increased too quickly?

A tolerability problem can sometimes be mistaken for an efficacy problem.

QUESTION 3

Has nutrition become the limiting factor?

Very low food intake, inadequate protein, dehydration, constipation, or declining activity can dramatically change how someone feels.

These are different problems.

They should not automatically receive the same solution.

Good tracking creates context.

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22 | WHAT WE KNOW

Human Phase 2 research supports several conclusions.

Eloralintide:

- selectively targets amylin receptor biology
- has been studied using once-weekly administration
- produced clinically meaningful average weight reduction
- demonstrated a dose-response relationship
- produced approximately 18–20% mean weight reduction in several higher-dose strategies at Week 48
- has now advanced into Phase 3 development

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23 | WHAT WE ARE LEARNING

Phase 3 is investigating:

- optimal dosing strategy
- maintenance exposure
- larger-population safety
- obesity with type 2 diabetes
- obesity without type 2 diabetes
- obesity-associated disease outcomes
- persistent obesity during incretin treatment
- amylin + incretin strategies
- longer-term outcomes

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24 | WHAT WE DO NOT KNOW YET

We do not currently have:

- completed Phase 3 efficacy results
- an FDA-approved eloralintide dose
- an FDA-approved titration schedule
- commercial prescribing information
- extensive long-term real-world safety data
- definitive head-to-head evidence against leading incretin therapies
- definitive evidence establishing the optimal amylin + incretin strategy

JOYOUS RULE

Investigational does not mean ineffective.

But it also does not mean established.

The evidence should be described according to where it actually sits.

25 | JOYOUS SCIENTIFIC INTELLIGENCE PANEL

Human Evidence

STRONG PHASE 2

Randomized, double-blind, placebo-controlled human evidence is available.

Clinical Development

PHASE 3 ACTIVE

Multiple large Phase 3 studies are underway.

Mechanistic Confidence

HIGH

The compound has a defined pharmacological target centered on selective amylin receptor agonism.

Dose Knowledge

DEVELOPING

Phase 2 dose-response information is available.

Final clinical dosing is not established.

Long-Term Evidence

DEVELOPING

Large-scale Phase 3 and eventual real-world evidence remain necessary.

Combination Potential

HIGH RESEARCH INTEREST

The use of amylin biology alongside incretin therapy is now being formally investigated.

26 | RESEARCH INTELLIGENCE SCORECARD

Mechanistic Evidence: ●●●●○

Human Clinical Evidence: ●●●●○

Dose-Response Evidence: ●●●●○

Long-Term Evidence: ●●○○○

Regulatory Maturity: ●●●○○

Combination-Therapy Interest: ●●●●●

Overall Research Position

Advanced Investigational

Eloralintide has progressed substantially beyond early experimental research but remains below the evidence threshold of an approved therapy with established prescribing information.

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27 | RUO DESCRIPTION

Eloralintide | LY3841136

Eloralintide, also identified as LY3841136, is an investigational long-acting selective amylin receptor agonist under clinical development for research involving metabolic signaling, satiety regulation, food intake, energy balance, and body-weight biology.

Human clinical research has evaluated once-weekly eloralintide across multiple dose levels and escalation strategies. A completed 48-week Phase 2 study investigated fixed 1 mg, 3 mg, 6 mg, and 9 mg research arms together with 6→9 mg and 3→6→9 mg escalation arms in adults with obesity or overweight without type 2 diabetes.

Eloralintide is scientifically distinct from GLP-1 receptor agonists because its primary pharmacological target is the amylin receptor system. Current clinical development is investigating both independent amylin signaling and its potential complementary role alongside incretin-based therapy.

Research Classification: Selective Amylin Receptor Agonist

Primary Research Areas: Amylin signaling • satiety biology • food-intake regulation • energy balance • obesity research • amylin-incretin interaction

Development Status: Phase 3 investigational compound

For Research Use Only. Not for human consumption or therapeutic use. Eloralintide is investigational and has not been approved by the U.S. Food and Drug Administration for the diagnosis, treatment, cure, or prevention of disease.

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28 | JOYOUS BOTTOM LINE

Eloralintide is important for a reason that extends beyond its Phase 2 weight-loss percentage.

It represents a broader shift in metabolic research.

The first major wave of modern obesity therapy demonstrated the power of incretin biology.

The next wave is asking:

What other biological systems can participate?

and eventually:

What happens when complementary systems are targeted together?

Eloralintide places **amylin signaling** squarely in that conversation.

Phase 2 established an important dose-response signal and demonstrated substantial average weight reduction across several dose strategies.

Phase 3 now has the harder job.

It must determine:

which dose, which escalation strategy, which population, which combinations, and at what balance of efficacy and tolerability.

That is the story to follow.

Not simply:

"How much weight did people lose?"

But:

"What is amylin biology going to add to the next generation of metabolic medicine?"

PRIMARY RESEARCH SOURCES

Phase 2 Clinical Trial
NCT06230523
48-week randomized, double-blind, placebo-controlled study of once-weekly eloralintide.

Phase 2 Publication
Lund and colleagues. *The Lancet*. 2025.
Eloralintide, a selective amylin receptor agonist for the treatment of obesity.

ENLIGHTEN-1
NCT07321886
Obesity / overweight without type 2 diabetes.

ENLIGHTEN-2
NCT07282600
Obesity / overweight with type 2 diabetes.

ENLIGHTEN-3
NCT07369011
Obesity / overweight with obstructive sleep apnea.

ENLIGHTEN-4
NCT07353931
Obesity / overweight with knee osteoarthritis pain.

ENLIGHTEN-6
NCT07392190
Persistent obesity during background weekly incretin therapy.

EDUCATIONAL & RESEARCH DISCLAIMER

This guide summarizes publicly available scientific and clinical-trial information for research education. Clinical-trial dose arms are presented to explain study design and should not be interpreted as prescribing instructions, individualized dosing recommendations, or self-administration protocols.

Eloralintide remains investigational and is not currently FDA approved.